

Linear Algebra and Calculus

SITAMS CSE (AI & ML) • R23 • I Year • I Semester • Course Code: 23BSC114T

Source basis: Official SITAMS R23 CSE (AI & ML) syllabus uploaded for this Study Hub. This concise revision guide preserves the syllabus topic statements and adds a study checklist; it is not a replacement for prescribed textbooks or faculty notes.

Unit I: MATRICES

Syllabus topics

- Rank of a matrix by echelon form, normal form
- Cauchy–Binet formulae (without proof)
- Inverse of Non- singular matrices by Gauss-Jordan method, System of linear equations: Solving system of Homogeneous and Non-Homogeneous equations by Gauss elimination method. Iterative methods: Jacobi and Gauss Seidel Methods

Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

Source page(s): 38

Unit II: EIGEN VALUES, EIGEN VECTORS AND ORTHOGONAL TRANSFORMATION

Syllabus topics

- Eigen values, Eigenvectors and their properties, Diagonalization of a matrix, Cayley- Hamilton Theorem (without proof), finding inverse and power of a matrix by Cayley- Hamilton Theorem, Quadratic forms and Nature of the Quadratic Forms, Reduction of Quadratic form to canonical forms by Orthogonal Transformation

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Unit III: CALCULUS

Syllabus topics

- Mean Value Theorems: Rolle's Theorem, Lagrange's mean value theorem with their geometrical interpretation, Cauchy's mean value theorem, Taylor's and Maclaurin's theorems with remainders (without proof), Problems and applications on the above theorems

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Unit IV: PARTIAL DIFFERENTIATION AND APPLICATIONS (MULTI VARIABLE CALCULUS)

Syllabus topics

- Functions of several variables: Continuity and Differentiability, Partial derivatives, total derivatives, chain rule, Taylor's and Maclaurin's series expansion of functions of two variables
- Jacobians, Functional dependence, maxima and minima of functions of two variables, method of Lagrange multipliers

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Unit V: MULTIPLE INTEGRALS(MULTIPLE VARIABLE CALCULUS)

Syllabus topics

- Double integrals, Triple integrals, change of order of integration, change of variables to polar, cylindrical and spherical coordinates
- Finding areas(by double integrals) and volumes (by double integrals and triple integrals)

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Study Hub status

This guide is syllabus-derived. External resources and quiz questions are tracked separately and are only labelled verified after source checking.